ADDISON W. MOTTO

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SUMMARY

Mechanical engineer experienced across the full hardware lifecycle (CAD, GD&T drawings, prototyping, and validation), seeking to bring a rigorous design-build-test approach to challenging problems.

EDUCATION

Cornell University
M.Eng., Systems Engineering | Expected December 2026
B.S., Mechanical Engineering | 2026 | GPA: 3.83 | Magna Cum Laude | Dean's List (All Semesters)

ENGINEERING EXPERIENCE

Helbling Laboratory, Cornell University | Graduate Researcher | *2026 - Present*
- Developing autonomous insect-scale robots to perform sustained intelligent operation in real-world environments.
- Design and develop a millimeter-scale deployable braking system for Gammabot, a sub-1-gram water-surface robot, integrating custom voice-coil actuator and flexure mechanism within tight mass and space constraints.
- Model actuator output and brake loading to size the actuator for sufficient force to retract the brake from the water; model hydrodynamic drag and deceleration, validating predictions through experimentation.
- Fabricate miniature electromagnetic and structural components using carbon-fiber laminate layup, laser micromachining, and precision assembly.

Creative Engineering LLC | Mechanical Design Intern | *Summer 2025*
- Intern at an engineering firm delivering innovative designs across a broad range of products.
- Delivered a 5-ft interactive wind turbine exhibit on schedule with client sign-off, owning mechanical development from SolidWorks design through vendor coordination, assembly, testing, and installation.
- Designed components for 6 client products, including the Times Square New Year's Eve Ball, applying GD&T and DFM/DFA across machined, welded, 3D-printed, laser-cut, and injection-molded parts.
- Performed statics and materials calculations to inform mechanical designs for patent-pending devices, manufacturing equipment, tooling, and a defense-related project.
- Presented CAD models and design rationale in recurring reviews with senior engineers and clients, incorporating feedback to refine designs against customer requirements.

Raytheon (RTX Corp) | Mechanical Engineering Intern | *Summer 2024*
- Intern in Rapid Hardware Solutions Department supporting the build effort for radars and advanced defense systems.
- Developed preset cable-cutting programs that reduced assembly time by 1-2% across a 20-technician wire harness shop; selected as 1 of 10 interns to present the project to engineering leadership.
- Supported radar and sensor hardware builds in Rapid Hardware Solutions, using Creo and SAP PRISM in a hands-on manufacturing environment spanning design, assembly, and quality assurance.
- Identified and corrected dozens of drawing, GD&T, and test-procedure issues before production.
- Tested hardware against DoD protocols and performed root-cause analysis to resolve product failures.
- Held a U.S. Secret security clearance during internship.

Hikma Pharmaceuticals | Manufacturing Intern | *Summer 2023*
- Provided hands-on support for automated, high-speed liquid-filling lines producing sterile injectables in an FDA- and DEA-regulated facility, following cGMP and standard operating procedures.

CORNELL HIGHLIGHTS

Cornell Electric Vehicles | Student Project Team | *2023 - 2025*
- Led design and fabrication of a dynamometer using load-cell torque and tachometer speed measurements to determine power; delivered under a tight deadline using primarily salvaged parts.
- Designed power-transmission and autonomous steering and braking mechanisms, integrating motors and sensors with electrical and software systems for the Shell Eco-Marathon.

Cornell Engineering Learning Initiatives | Tutor | *2023 - 2024*
- Recruited by Cornell to tutor peers in Physics, Multivariable Calculus, and Chemistry.

TECHNICAL SKILLS

CAD and analysis: SolidWorks, Creo, Onshape, Fusion 360, Autodesk Inventor; Ansys Mechanical, Fluent, and Thermal; MATLAB, Python, LabVIEW, Arduino programming.
Design and manufacturing:GD&T; DFM/DFA; 3D printing; laser micromachining; carbon-fiber fabrication.